

# Module 10: Epigenetics — Keys to Success

## Learning Objectives

By the end of this module, you should be able to:

1. Explain why cells with the same DNA can look and act differently
2. Define epigenetics and explain how it differs from a mutation
3. Describe how DNA methylation and histone modification control gene activity
4. Give examples of how the environment can affect which genes are turned on or off

## The Big Picture

Every cell in your body has the **same DNA**. A skin cell and a brain cell are genetically identical, yet they look and function completely differently. **Gene regulation** controls which genes are "on" in each cell type. **Epigenetics** is how these on/off patterns are set without changing the DNA itself. Think of it this way: your DNA is the recipe book, and epigenetics decides which recipes get cooked.

## Key Terms

### Gene Regulation — The Basics

- **Gene Expression** — Using a gene's instructions to make a protein; a gene that is being used is said to be "expressed"
- **Gene Regulation** — The process of turning genes on or off; this is how one set of DNA can produce many different cell types
- **Transcription Factor** — A protein that helps turn a gene on or off by binding near the gene
- **Promoter** — The specific spot on DNA where the cell's machinery attaches to begin reading a gene

## Epigenetics — Changing Gene Activity Without Changing DNA

- **Epigenetics** — Changes in gene activity that do NOT change the DNA sequence itself
- **DNA Methylation** — Small chemical tags (methyl groups) are added to DNA → the gene is turned **OFF** (silenced)
- **Histone Modification** — Chemical tags are added to histone proteins (the "spools" DNA wraps around):
  - Acetylation → DNA loosens → gene is easier to read → gene **ON**
  - Methylation of histones → DNA tightens → gene harder to read → gene **OFF**
- **Key point:** Epigenetic changes are **reversible** — they can be undone, unlike permanent mutations

## X-Inactivation — A Visible Example

- In females (XX), one X chromosome in each cell is randomly shut down early in development → forms a **Barr body**
- **Calico cats** — The patchy coat pattern is a visible result of random X-inactivation in different skin cells
- This is one of the most dramatic everyday examples of epigenetics

## How the Environment Affects Your Genes

| Factor             | How It Can Change Gene Activity                                                  |
|--------------------|----------------------------------------------------------------------------------|
| Diet               | Nutrients like folic acid provide methyl groups that affect methylation patterns |
| Stress             | Chronic stress can change which genes are active, especially in the brain        |
| Exercise           | Regular physical activity can turn on genes involved in metabolism and health    |
| Toxins / Chemicals | Pollutants and drugs can disrupt normal epigenetic patterns                      |

**Your genes are not your destiny.** Your lifestyle and environment influence which genes are turned on or off throughout your life.

## Study Tips

1. **Same DNA, different cells** — This is THE core concept; if you understand this, you understand the module
2. **Epigenetics  $\neq$  mutation** — Epigenetics changes the *volume* of a gene, not the *song*; mutations change the song itself
3. **Methylation = Muting** — Both start with "M"; methylation silences genes
4. **Acetylation = Activation** — Both start with "A"; acetylation opens up DNA for reading
5. **Think about calico cats** — The single best real-world example of epigenetics you can point to